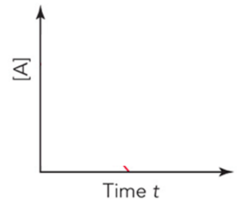
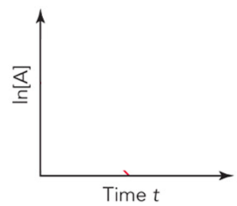
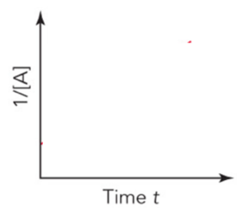
Exam 3 Review worksheet

Fill out the following review table.

Order	Rate Law	Units of *k*	Integrated Rate Law	Straight-Line Plot	Half-Life Expression
0	Rate =				
1	Rate =				
2	Rate =				

1) Write a balanced reaction for which the following rate relationships are true.

$$\text{Rate} = \frac{1}{2}\frac{\Delta[N_2]}{\Delta t} = \frac{\Delta[O_2]}{\Delta t} = -\frac{1}{2}\frac{\Delta[N_2O]}{\Delta t}$$

A) $\frac{1}{2} N_2 + O_2 \rightarrow \frac{1}{2} N_2O$

B) $2 N_2O \rightarrow 2 N_2 + O_2$

C) $N_2O \rightarrow N_2 + 2 O_2$

D) $\frac{1}{2} N_2O \rightarrow \frac{1}{2} N_2 + O_2$

E) $2 N_2 + O_2 \rightarrow 2 N_2O$

2) Use the information in Q1 for this question, The change in concentration of O_2 over time is 10. M/s, what is the change in concentration of N_2O over time?

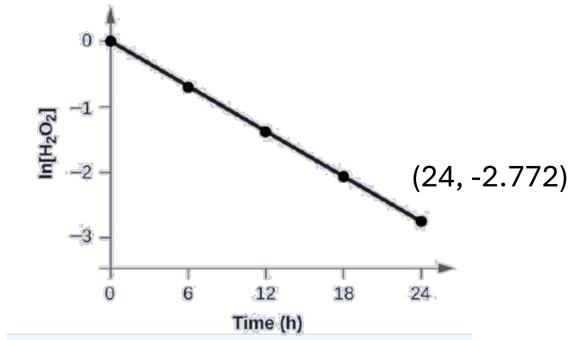
3) A rate is equal to 0.0200 M/s. If [A] = 0.100 M and rate = $k[A]^2$, what is the new rate if the concentration of [A] is increased to 0.200 M?

4) Determine the rate law for the following reaction using the data provided.

$2\,NO(g) + O_2(g) \rightarrow 2\,NO_2(g)$	$[NO]_i$ (M)	$[O_2]_i$ (M)	Initial Rate (M/s)
	0.030	0.0055	8.55×10^{-3}
	0.030	0.0110	1.71×10^{-2}
	0.060	0.0055	3.42×10^{-2}

5) The rate constant for the first-order decomposition of C_4H_8 at 500 °C is 9.2×10^{-3} s^{-1}. How long will it take for 80.0% of a sample of C_4H_8 to decompose?

6) The data from this Figure with the addition of value of $\ln[H_2O_2]$ are given in Figure below.



a. What is the rate law for this reaction?

b. What is the concentration of H_2O_2 after two day?

c. How long does it take for 75% of H_2O_2 to react?

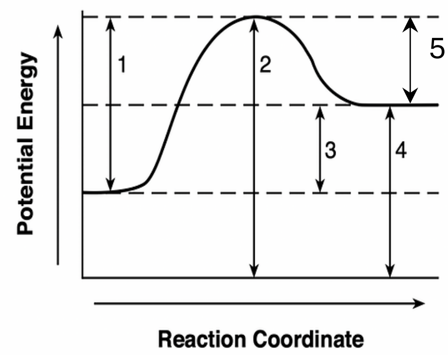
7) The activation energy of a reaction is 73.11 kJ/mol, and the frequency factor is 1.3×10^{10} $M^{-1}s^{-1}$. Knowing that it is a second order reaction, what is the initial rate of the reaction with an initial concentration of 0.100 M at 40°C.

8) Given the potential energy diagram for a reaction on the right.

a. Label where is the reactant, product, and transition state.

b. which number represent the activation energy for the forward reaction?

c. which number represent the activation energy for the reversed reaction?



d. Which number represent the enthalpy of the reaction?

e. Is the forward reaction an exothermic or endothermic reaction?

9) Determine the overall reaction and the rate law for the mechanism below.

Step 1: $H_2O_2(aq) + I^-(aq) \rightarrow H_2O(l) + IO^-(aq)$ slow

Step 2: $H_2O_2(aq) + IO^-(aq) \rightarrow H_2O(l) + O_2(g) + I^-(aq)$ fast

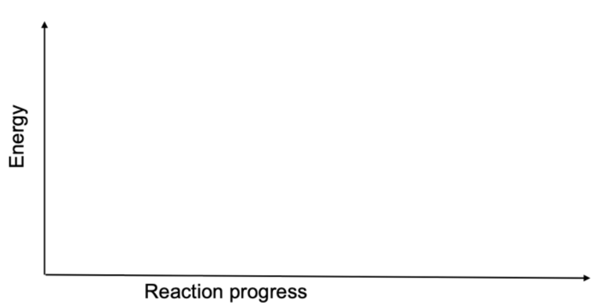
a. What is the overall reaction?

b. Which step is rate determining step?

c. Identify the intermediate(s) of this reaction and decide if the intermediate is catalyst or not.

d. What is the expected order of the reaction based on the proposed mechanism?

e. Draw the reaction coordinate diagram and label the position of reactants, intermediates and products. Knowing that the reaction enthalpy is -50kJ/mol, E_a for rate determining step is 150 kJ/mol while the E_a for the other step is 50 kJ/mol.



10) Consider the reaction below. At equilibrium at 298 K, the reaction contains 0.12 mol of $Pb(NO_3)_2$, 0.15 mol of KCl, 1.23 mol of $PbCl_2$, and 3.14 mol of KNO_3 in a 1.00 L of reaction vessel. Calculate the rate constant for the reaction at 298 K.

$$Pb(NO_3)_2(aq) + 2\,KCl(aq) \rightleftharpoons PbCl_2(s) + 2\,KNO_3(aq)$$

11) Use the equilibrium constant given below to determine equilibrium constant for the following reaction: $P_4(g) + 10\,Cl_2(g) \rightarrow 4\,PCl_5(s)$

Given:

$PCl_5(s) \rightarrow PCl_3(g) + Cl_2(g)$ $K = 2.5*10^{-2}$

$P_4(g) + 6\,Cl_2(g) \rightarrow 4\,PCl_3(g)$ $K = 6.2*10^{3}$

At the equilibrium, does the reaction mixture favors reactants or products?

Useful Equations:

$[A]_t = [A]_0 e^{-kt}$ $\quad$ $\ln[A]_t = \ln[A]_0 - kt$ $\quad$ $[A]_t = [A]_0 - kt$ $\quad$ $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$

$t_{1/2} = \frac{[A]_0}{2k}$ $\quad$ $t_{1/2} = \frac{1}{k[A]_0}$ $\quad$ $t_{1/2} = \frac{\ln 2}{k}$

$k = Ae^{-\frac{E_a}{RT}}$

R = 8.314 J/(mol•K)

Answer: Q1) B; Q2) -20. M/s; Q3) 0.0800 M/s; Q4) Rate = 1.7×10^3 $M^{-2}s^{-1}$ $[NO]^2[O_2]$; Q5) 170 s; Q6) a: r=0.1155 $h^{-1}[H_2O_2]$, b: $3.91*10^{-3}$M, c: 12 h, Q7) 8.3×10^{-5} M/s; Q8) b:1, c: 5, d: 3, e: endothermic; 9) a: $2H_2O_2(aq) \rightarrow 2H_2O(l) + O_2(g)$, b: step 1, c: I^- (catalyst), IO^-, b:2; 10) 3652; 11) $1.6*10^{10}$, products